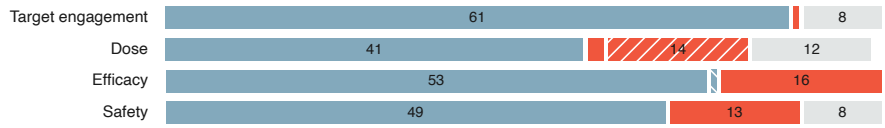
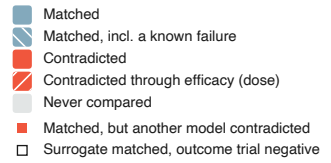
a





Panel b: every drafted tile; diseases by global DALYs in 2021 (WHO Global Health Estimates), then those without a WHO cause by area.

b

		Target engagement	Dose	Efficacy	Safety
#1	**type 2 diabetes mellitus** GLP1R agonists (12)	**2011** Cynomolgus monkeys and rat islets (dulaglutide)	**2019** Beagle dog oral bioavailability of semaglutide/SNAC tablets	**2023** GLP1R Gly168Ser genotype, GLP-1 infusion clamp in healthy volunteers	**2016** GLP1R Ala316Thr carriers (human genetics)
	type 2 diabetes mellitus GCK activators (9)	**2011** Rodent models of type 2 diabetes	**2012** Glucose-insulin-glucagon model (LY2599506) Dose reached; effect faded	**2019** People with GCK-activating or GKRP loss-of-function variants AZD1656 missed 4-month HbA1c	**2024** Rodent models of type 2 diabetes GCK proxy misses GKA lipid rise
	type 2 diabetes mellitus HSD11B1 inhibitors (8)	**2022** Diet-induced obese and KKAy mice dosed with a selective… Adipose inhibition held 14 days;…	**2022** Diet-induced obese and KK… Full inhibition reached; no glucose effect	**2022** Diet-induced obese and KK… Complete inhibition, no glucose lowering	**2022** Cortisone reductase deficiency: people with loss-of-function HSD11B1 or H6P…
	type 2 diabetes mellitus DPP4 inhibitors (7)	**2016** Rodent plasma DPP-4 inhibition and oral glucose/carbohydrate load (Wistar…	**2016** Rodent plasma DPP-4 inhibition and oral glucose/carbohydrate load (Wistar…	**2023** Population PK/PD plus model-based meta-analysis (cofrogliptin single-ascending-dose…	**2016** Concentration-QTc model from the omarigliptin… Monkey skin harm absent in…
	type 2 diabetes mellitus INSR agonists (6)	**2023** STZ-diabetic rat single-dose PK/glucodynamics (Lilly)	**2026** LYD pig euglycaemic clamp + rat/dog PK	**2022** STZ-diabetic rat single-dose PK/glucodynamics (Lilly)	**2015** In vitro mitogenicity in L6-hIR, HMEC and Saos-2 cells Liver fat, ALT rise missed in…
	type 2 diabetes mellitus GCGR/GLP1R agonists (5)	**2023** Human GLP-1 + glucagon co-infusion challenge (overweight volunteers without diabetes)	**2020** Diet-induced obese (DIO)… GI intolerance capped dose below glucagon engagement	**2021** Diabetic db/db mouse (chronic glucose lowering, Sanofi exendin-based dual agonists)	**2020** Healthy cynomolgus monkey (4- and 8-week repeat-dose… Vomiting 17 % in patients
	type 2 diabetes mellitus SLC5A2 inhibitors (5)	**2012** ZDF rat renal glucose threshold	**2013** ZDF rat efficacy exposure with SD-rat/dog toxicology margins	**2023** db/db mouse and ZDF rat 4-week HbA1c	**2023** Rat 2-year carcinogenicity and chronic toxicology (SD rat, CD-1 mouse, dog)
	type 2 diabetes mellitus GCGR antagonists (4)	**2015** Anti-GCGR antibody in ob/ob and C57BL/6 mice and cynomolgus monkeys	**2016** Glucagon-challenge hGCGR mice and rhesus monkeys; diabetic mice (small-molecule GRAs)	**2015** Gcgr-/- knockout mouse (genetic full antagonism)	**2022** Anti-GCGR antibody in ob/o… Transaminase, LDL-C, BP rises missed (T1D)
	type 2 diabetes mellitus GPR119 agonists (4)	**2014** Human primary colonic L-cell cultures (GLP-1/PYY secretion ex… GLP-1/GIP/insulin flat in T2D;…	**2014** GPR119-agonist-dosed mice (OGTT; insulin, GLP-1/GIP release)	**2018** GPR119-agonist-dosed mic… Firuglipel failed HbA1c; comparator worked	**2018** GPR119-agonist-dosed mice (OGTT; insulin, GLP-1/GIP release)
	type 2 diabetes mellitus CCR2 antagonists (3)	**2013** Diet-induced obese / db/db mice dosed with a CCR2 antagonist	**2013** Human-CCR2 knock-in DIO / db/db mice dosed with CCX140-B	**2013** Diet-induced obese / db/db mice dosed with a CCR2 antagonist	**2005** Diet-induced obese / db/db mice dosed with a CCR2 antagonist
#2	**chronic obstructive pulmonary disease** CHRM3 antagonists (6)	**2018** Human isolated bronchial tissue: carbachol-evoked contraction, potency and washout…	**2018** Human isolated bronchial tissue: carbachol-evoked contraction, potency and washout…	**2018** Human isolated bronchial tissue: carbachol-evoked contraction, potency and washout…	**2018** Rat methacholine bronchoprotection (Einthoven) and… Program stopped; preclinical dat…
	chronic obstructive pulmonary disease ADRB2 agonists (4)	**2016** Almirall guinea-pig acetylcholine bronchoprotection, human isolated bronchi, and dog…	**2016** Almirall guinea-pig acetylcholine bronchoprotection, human isolated bronchi, and dog…	**2016** Almirall guinea-pig acetylcholine bronchoprotection,… ADRB2 genotype did not predict…	**2016** Almirall guinea-pig acetylcholine bronchoprotection, human isolated bronchi, and dog…
	chronic obstructive pulmonary disease MAPK14 inhibitors (4)	**2018** AstraZeneca human alveolar-macrophage/whole-blood LPS-TNF potency model with rat…	**2020** Acumapimod PBPK drug-drug-interaction model… Lung target hit; 1 mg trial failed	**2019** Rodent inhaled-LPS airway… Acumapimod missed FEV1 primary twice	**2018** AstraZeneca human alveolar-macrophage/whole-blood LPS-TNF potency model with rat…
#3	**HIV** HIV RT (NNRTI) inhibitors (3)	**2012** HIV-1 replication in cell culture (MT-4/PBMC) against wild-type and NNRTI-resistant viru…	**2020** In-silico doravirine PK model for CYP3A-inducer co-administration: nonparametric…	**2012** HIV-1 replication in cell culture (MT-4/PBMC) against wild-type and NNRTI-resistant virus	**2012** Nonclinical toxicology in standard species (rodent plus… Seizures no tox species predicted
	HIV HIV RT (NRTI) inhibitors (3)	**2013** HIV-1 replication in cell culture (CEM/PBMC/MT-4) with site-directed NRTI-resistant mutant…	**2021** Cell-culture potency plus… Threefold escalation still failed; abandoned	**2021** HIV-1 replication in cell cultu… Resistant-virus activity failed in patients	**2015** Chronic rat and monkey toxicology (6- and 9-month studies)
	HIV HIV integrase (INSTI) inhibitors (3)	**2011** HIV-1 replication in cell culture (MT-4/PBMC)	**2013** Protein-adjusted cell-culture potency as plasma trough target	**2012** In-vitro susceptibility of INSTI-resistant HIV-1 (site-directed mutant clones, fold-change)	**2026** Rat and rabbit embryo-fetal development studies Mouse NTD; human signal gone
	HIV HIV maturation (Gag) inhibitors (3)	**2021** HIV-1 replication in cell culture against patient-derived and site-directed Gag-polymorphic…	**2023** Population PK + sigmoid Emax exposure-response model fitted to Phase 2a monotherapy…	**2023** HIV-1 replication in cell culture against patient-derived and site-directed Gag-polymorphic…	**2025** Nonclinical toxicology (short… Animal gastric lesions absent in patient biopsies
#4	**NSCLC** EGFR inhibitors (18)	**2017** T790M/sensitising-mutant NSCLC cells and mice	**2019** PC9 brain-metastasis mice; monkey osimertinib PET	**2023** Patient-derived lung cancer organoids EXCLAIM-2 fails primary PFS	**2015** hERG, rat and dog tox, dog telemetry QT prolongation the package cleared
	NSCLC ALK inhibitors (6)	**2019** Ba/F3 EML4-ALK L1196M cells and crizotinib-resistant patient-derived lines	**2018** EML4-ALK NSCLC xenografts (H2228/H3122, subcutaneous and intracranial)	**2019** Ba/F3 EML4-ALK L1196M cells and crizotinib-resistant patient-derived lines	**2018** Rat, monkey and dog toxicology
	NSCLC MET inhibitors (5)	**2021** MET-driven human tumour xenografts/PDX (METex14, high-level MET-amplified) and…	**2021** MET-driven human tumour xenografts/PDX (METex14, high-level MET-amplified) and…	**2023** Patient-derived lung cancer organoids from malignant effusions or tissue, drug-sensitivity tested…	**2021** GLP rat (26-week) and beagle dog (39-week) toxicology;… Oedema and ILD unpredicted;…
	NSCLC anti-IGF1R antibodies (4)	**2011** Tumour cells and xenografts given h7C10	**2005** A549 NSCLC xenografts in nude mice	**2011** A549 NSCLC xenografts in nude mice R1507 + erlotinib: no PFS gain	**2021** Cynomolgus repeat-dose tox (R1507) Monkeys missed hyperglycemia
	NSCLC anti-PDCD1 antibodies (4)	**2023** NSG mice reconstituted with human PBMCs, with T-cell expansion and effector-memory…	**2023** Cynomolgus monkey pharmacology and repeat-dose toxicology with a PD-1…	**2023** NSG mice bearing MC38 tumours engrafted with allogeneic human CD8+ cytotoxic T…	**2021** PD-1 knockout mice Monkeys silent; patients had irAEs
	NSCLC pan-HER inhibitors (4)	**2011** EGFR del19/L858R NSCLC lines (HCC827, HCC4006) and HCC827 xenografts	**2017** EGFR del19/L858R NSCLC lines (HCC827, HCC4006) and… Stated 16 mg margin collapsed t…	**2017** EGFR del19/L858R NSCLC lines (HCC827, HCC4006) and… Registrational exon-20 cohort…	**2020** Hypoxia-activated prodrug in EGFR exon 20 / ERBB2-mutant… PDX silent on 20% grade 3…
	NSCLC HSP90AA1/HSP90AB1 inhibitors (3)	**2013** Genotype-defined NSCLC cell lines and xenografts (H1975 and others)	**2013** Genotype-defined NSCLC c… Exposure reached; the class still failed	**2018** EGFR exon 20 insertion cell models (Ba/F3 and NSCLC lines) Phase 3 with docetaxel failed: O…	**2013** Mouse xenograft studies with cardiac and liver readouts (ganetespib)
	NSCLC VEGFR multikinase inhibitors (3)	**2022** Syngeneic Lewis lung carcinoma (LLC) subcutaneous tumours in immunocompetent…	**2022** Syngeneic Lewis lung carcinoma (LLC) subcutaneous tumours in immunocompetent…	**2019** RET fusion-driven cells (CCDC6-RET lung and thyroid line… Three LEAP phase 3s failed; O…	**2019** GLP repeat-dose toxicology in rat plus monkey (and dog)
	NSCLC anti-CTLA4 antibodies (3)	**2010** B16, MC38 and CT26 tumours in mice with anti-mouse CTLA-4 plus anti-PD-1	**2019** B16, MC38 and CT26 tumours in mice with anti-mouse… 10 mg/kg reached; failed;…	**2020** B16, MC38 and CT26 tumours in mice with anti-mouse… Nivo-ipi-chemo OS met, HR 0.69	**2011** CTLA-4 knockout mice Monkeys missed fatal immune…
#5	**migraine disorder** CGRP receptor antagonists (4)	**2014** Rhesus capsaicin-induced dermal vasodilation (CIDV)	**2014** Rhesus CIDV PK/PD target Predicted 20 mg; needed 200 mg	**2013** Primate CGRP-induced facial blood flow (rimegepant / BMS-927711)	**2026** Mouse MCAO stroke, gepant pretreatment Animal tox missed MK-3207 liver
	migraine disorder anti-CALCA/CALCB (CGRP ligand)…	**2015** Cynomolgus capsaicin-induced dermal blood flow (CIDBF), PK/PD-translated to…	**2015** Cynomolgus capsaicin-induced dermal blood flow (CIDBF), PK/PD-translated to…	**2014** Rat evoked neurogenic vasodilation (skin flare, middle meningeal artery)	**2025** Rat chronic blood pressure and heart rate under anti-CGRP… Postmarketing hypertension warning
#6	**Alzheimer disease** BACE1 inhibitors (4)	**2015** PDAPP mice and cannulated beagle dogs (brain/CSF Aβ)	**2019** Dog CSF Aβ PK/PD scaled to humans Target reached, no benefit	**2019** A673T variant carriers Genetics said protect; drug worsened	**2019** Mice chronically dosed with BACE1 inhibitors Liver injury missed by tox
	Alzheimer disease anti-MAPT antibodies (4)	**2023** Tau-transgenic mouse interstitial fluid (gosuranemab) CSF Abeta42 unchanged despit…	**2023** Tau-transgenic mouse… ISF target met in CSF; CDR-SB unchanged	**2022** Human AD-brain PHF-tau seeding assay (HEK293 P301S-ta… AD-seed removal did not translate	**2021** Cynomolgus monkeys and tau-transgenic mice (PK/PD; toxicology species unpublished)